

```
from pqcrypto.sign import sphincsplus

sk, pk = sphincsplus.generate_keypair()

message = b"Secure blockchain transaction"

signature = sphincsplus.sign(message, sk)

assert sphincsplus.verify(message, signature, pk)

print("Signature verified successfully!")
```

Multivariate polynomial

cryptography: This relies on factoring hard multivariate quadratic equations and is thus quantum resistant. Techniques such as Rainbow offer secure digital signatures to be used in smart contracts, transaction signing, and decentralised identity management.

Code-based cryptography:

Code-based cryptography, such as Classic McEliece, employs error-correcting codes to offer a mathematically established model of security against quantum attackers. This is appropriate for long-term data encryption in blockchain, wallet security, and communication between blockchain nodes.

Super singular isogeny-based

cryptography: This is a new cryptographic system that relies on isogeny computation hardness between super singular elliptic curves. While still in the research phase, SIKE (super singular isogeny key encapsulation) provides a lightweight option for low-power blockchain devices, IoT-based blockchain applications, and distributed networks.

The US-based National Institute of Standards and Technology (NIST) is currently standardising PQC algorithms for global use. Blockchain developers are also seeking ways to implement such cryptographic methods in existing and future blockchain platforms.

Integrating AI with quantum-resistant blockchains

AI and PQC integration strengthen blockchain resilience, efficiency, and quantum attack immunity. AI-based techniques improve anomaly detection, cryptographic processing, and scalability.

AI-based anomaly detection: AI models track blockchain transactions to identify anomalies and thwart quantum-driven cyberattacks.

Here is an example of how AI can help identify suspicious blockchain transactions:

```
from sklearn.ensemble import IsolationForest

import numpy as np

data = np.random.rand(100, 5)

model = IsolationForest(
    contamination=0.05).fit(data)

anomalies = model.predict(data)

print("Anomalous Transactions:",
      np.where(anomalies == -1))
```

AI for blockchain scalability: AI reduces blockchain operation latency by predicting network congestion and expanding transaction capacity. Efficient blockchains are enabled by

AI-based consensus models, while PQC implementations offer robust efficiency, and secure and quick transactions.

AI-driven quantum-resistant blockchains can enable secure, efficient, and scalable cryptographic transactions in a decentralised and quantum-immune Web3 society.

The future of quantum-resistant AI blockchains

It is imperative that developers and researchers already engaged in developing blockchain technology actively adopt quantum-resistant methods. Governments and institutions are racing towards standardising and implementing quantum-resistant cryptographic protocols. In the meantime, pre-quantum and post-quantum cryptography can be hybridized as an intermediate step to ensure backward compatibility.

Quantum computing is both an opportunity and challenge for the blockchain ecosystem. By adopting quantum-resistant cryptography and AI-based security upgrades, blockchain networks can maintain their development trajectory and protect the decentralised Web3 future. Integrating these pioneering technologies will ensure secure and reliable blockchain ecosystems in a post-quantum world. **END** 🐧

References

- Boneh, D., & Lipton, R. J. (1995), Quantum cryptanalysis of hidden linear functions, *Advances in Cryptology – CRYPTO '95*, 424–437
- National Institute of Standards and Technology (NIST) (2022), Post-Quantum Cryptography Standardization, <https://csrc.nist.gov/projects/post-quantum-cryptography>
- <https://www.qnulabs.com/blog/post-quantum-cryptography-the-complete-guide>
- <https://www.mdpi.com/1424-8220/23/3/1228>
- Kiktenko, E. O., Trushechkin, A.S., Yunusov, R. R., et al (2018), Quantum-secured blockchain technology, *Quantum Science and Technology*, 3(3), 035004

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